

1. Universal Proof

- Using predicate logic, prove: $\forall n \in \mathbb{Z}$, if n is even then n^2 is even.
- Prove: $\forall n \in \mathbb{Z}$, $n(n+1)$ is even.
- Prove: $\forall x \in \mathbb{R}$, $x^2 \geq 0$.

2. Existential Proof

- Prove: $\exists n \in \mathbb{Z}$ such that n is even and divisible by 6.

Prf:

Choose $n=12$. Then $n \in \mathbb{Z}$ and $6|12$.

- Prove: $\exists x \in \mathbb{R}$ such that $x^2 = 25$.

Prf: choose $x=5$.

- Prove: $\exists n \in \mathbb{Z}$ such that n is odd and n^2 is odd.

Then $n \in \mathbb{Z}$ and $n^2 = 25$.

Choose $n=3$ - odd
 $n^2=9$ - odd

3. Direct Proof

- Prove directly: $\forall n \in \mathbb{Z}$, if 4 divides n , then 2 divides n .
- Prove directly: $\forall a, b \in \mathbb{Z}$, if a and b are odd then ab is odd.
- Prove directly: $\forall x \in \mathbb{R}$, if $x > 2$ then $x^2 > 4$.

Let $n \in \mathbb{Z}$ — (i)
Suppose $4|n$ — (ii)

$\therefore 2|n$ — (iii)
By (i), (ii) & (iii) $\forall n \in \mathbb{Z}$, $4|n \Rightarrow 2|n$. $\square$

4. Proof by Cases

- Using cases, prove: $\forall n \in \mathbb{Z}$, $n^2 + n$ is even.
- Prove by cases that $|x| \geq 0$ for all real numbers x .
- Using even/odd cases, prove: $\forall n \in \mathbb{Z}$, $n^2 - n$ is even.

Let $n \in \mathbb{Z}$
Case (i) n is even
Case (ii) n is odd

$\forall x \in D, p(x) \Rightarrow q(x) \equiv \underline{\underline{\forall x \in D, \neg p(x) \Rightarrow \neg q(x)}}$

5. Proof by Contraposition

- Prove by contraposition: $\forall n \in \mathbb{Z}$, if n^2 is odd then n is odd.
- Prove by contraposition: if ab is even, then a is even or b is even.
- Prove by contraposition: if $x^2 \geq 9$, then $x \geq 3$ or $x \leq -3$.

Prf: Let $n \in \mathbb{Z}$
Suppose n is even
 $\vdots$
 n^2 is even

6. Proof by Contradiction

(4) " $\forall n \in \mathbb{Z}, n^2 \text{ is odd} \Rightarrow n \text{ is odd}$ "

- Prove by contradiction that there is no integer that is both even and odd.
- Prove by contradiction that $\sqrt{2}$ is irrational.
- Prove by contradiction: there is no smallest positive rational number.

Prf: Let $\forall n \in \mathbb{Z}, n^2 \text{ is odd} \Rightarrow n \text{ is odd}$ $\Leftarrow$ (i)
Assume that (i) is false
Then $\exists n \in \mathbb{Z}$ such that n^2 is odd and n is even
 $\vdots$
 $\therefore$ a contradiction

7. Proof of If and Only If

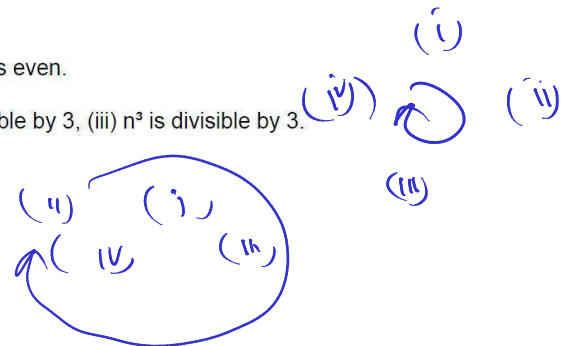
1. Prove: an integer n is even if and only if n^2 is even.
2. Prove: n is divisible by 6 if and only if n is divisible by both 2 and 3.
3. Prove: $|x| < a$ if and only if $-a < x < a$, where $a > 0$.

$(i) \quad n \text{ is even} \quad p$
 $(ii) \quad n^2 \text{ is even} \quad q$
 Proof:
 proof of $(i) \Rightarrow (ii) \quad (p \Rightarrow q)$
 proof of $(ii) \Rightarrow (i) \quad (q \Rightarrow p) \equiv (\neg p \Rightarrow \neg q)$

8. Proof of Equivalence of Three Statements

1. Show that the following are equivalent for $n \in \mathbb{Z}$: (i) n is even, (ii) n^2 is even, (iii) n^3 is even.
2. Show that the following are equivalent for $n \in \mathbb{Z}$: (i) n is divisible by 3, (ii) n^2 is divisible by 3, (iii) n^3 is divisible by 3.

$(i) \Rightarrow (ii)$
 $(ii) \Rightarrow (i)$
 $(ii) \Rightarrow (i)$



Hints for Mid-Exam I

01. Formulate an argument and determine its validity without using a truth table.

02. You should be able to give:

- | | |
|--|------------------------------|
| (i) Universal proofs | (v) Proofs by contraposition |
| (ii) Existential and uniqueness proofs | (vi) Proofs by contradiction |
| (iii) Direct proofs | (vii) If and only if proofs |
| (iv) Proofs by cases | (viii) Proofs of equivalence |

Proof by Principle of Mathematical Induction

The principle of Mathematical Induction:

For each positive integer n , let $P(n)$ be a statement.

If

- (1) $P(1)$ is true and
 - (2) If $P(k)$, then $P(k+1)$ is true for every positive integer k ,
- then, $P(n)$ is true for every positive integer n .

More symbolically, the principle of Mathematical Induction can be stated as:

For each positive integer n , let $P(n)$ be a statement.

If

- (1) $P(1)$ is true and
 - (2) $\forall k \in \mathbb{N}, P(k) \Rightarrow P(k+1)$ is true,

then, $\forall n \in \mathbb{N}, P(n)$ is true.

Principle of Induction. In order to show that $\forall n, P(n)$ holds, it suffices to establish the following two properties:

- (1) **Base case:** Show that $P(1)$ holds.
- (2) **Induction step:** Assume that $P(k)$ holds, and show that $P(k + 1)$ also holds.

In the induction step, the assumption that $P(k)$ holds is called the **Induction Hypothesis**.

In more formal notation, this proof technique can be stated as

$$[P(1) \wedge \forall k (P(k) \Rightarrow P(k + 1))] \Rightarrow \forall n P(n)$$

Note:

Principle of Mathematical Induction

The primary use of the Principle of Mathematical Induction is to prove statements of the form

$$\forall n \in \mathbb{N}, P(n)$$

Procedure for a Proof by Mathematical Induction

To prove: $\forall n \in \mathbb{N}, P(n)$

Basis step: Prove $P(1)$.

Inductive step: Prove that for each $k \in \mathbb{N}$,
if $P(k)$ is true, then $P(k + 1)$ is true.

We can then conclude that $P(n)$ is true for all $n \in \mathbb{N}$.

Example 1. If $n \in \mathbb{N}$, then $1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$.

i.e. Prove that $\forall n \in \mathbb{N}, 1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$.

Example 2. If n is a non-negative integer, then $5 \mid (n^5 - n)$.

i.e. Prove that $\forall n \in \mathbb{N} \cup \{0\}, 5 \mid (n^5 - n)$.

Ex. The inequality $2^n \leq 2^{n+1} - 2^{n-1} - 1$ holds for each $n \in \mathbb{N}$.

$$\forall n \in \mathbb{N}, 2^n \leq 2^{n+1} - 2^{n-1} - 1$$

Exercise

Let a_n be defined by $a_1 = \sqrt{2}$ and $a_{n+1} = \sqrt{2 + a_n}$ for $n \in \mathbb{N}$.

(a) Show that $a_n > 0$ for all $n \in \mathbb{N}$.

(b) Show that $a_n \leq 2$ for all $n \in \mathbb{N}$.

(c) Show that $a_{n+1} \geq a_n$ for all $n \in \mathbb{N}$.

Proof by Strong Induction

Principle of Strong Mathematical Induction: To prove that $P(n)$ is true for all positive integers n , we complete these steps:

- *Basis Step:* Show that $P(1)$ is true.
- *Inductive Step:* Show that $\forall k \in \mathbb{N}, [P(1) \wedge P(2) \dots \wedge P(k)] \Rightarrow P(k + 1)$
- Inductive Hypothesis includes ALL k statements $P(1), P(2), \dots, P(k)$.

Example 1: Given $n \in \mathbb{N}$, define a_n recursively as follows:

$$a_0 = 1, a_1 = 3, a_n = 2a_{n-1} - a_{n-2} \text{ for all } n \geq 2.$$

Prove that for all $n \in \mathbb{N}$, $a_n = 2n + 1$.

Example 2. Define a sequence recursively by:

$$a_n = \begin{cases} a_1 = 1 \\ a_2 = 4 \\ a_n = 2a_{n-1} - a_{n-2} \quad \text{For } n \geq 3 \end{cases}$$

Let's prove that $a_n = 3n - 2$ for all $n \geq 1$.

(a) The Base Case(s):

When proving the statement for $k + 1$ we had to use the statement for $k - 1$. Because our claim starts with $n = 1$ we have to ensure that $k - 1 \geq 1$, so $k \geq 2$.

This means that there are two base cases, a_1 and a_2 . We check those:

$a_1 = 1 = 3(1) - 2$ is true and $a_2 = 4 = 3(2) - 2$ is true.

(b) The Inductive Step:

We assume that $a_i = 3i - 2$ for $i = 1, 2, \dots, k$ (the induction hypothesis).

Observe that:

$$\begin{aligned} a_{k+1} &= 2a_k - a_{k-1} \\ &= 2(3k - 2) - (3(k - 1) - 2) \\ &= 6k - 4 - 3k + 3 + 2 \\ &= 3k + 1 \\ &= 3(k + 1) - 2 \end{aligned}$$

Then we prove that $a_{k+1} = 3(k + 1) - 2$

Example 3. Every positive integer greater than 1 can be expressed as a product of primes.

Ex. Let $f_n = f_{n-1} + f_{n-2}$, $f_0 = 1$, and $f_1 = 1$. Show that $f_n \leq 2^n$ for all positive n .

Ex. 1 of D2